

Trainings ▾

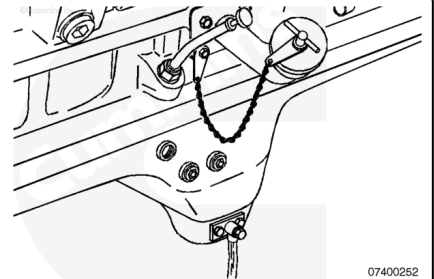
Drain

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Do not extend oil change intervals past 250 hours or 6 months, whichever comes first, unless the chart method is being used.

Operate the engine until the water temperature reaches 60°C [140°F].

Turn the engine off.

Remove the oil drain plug and open the drain valve.

Drain the oil immediately to be sure all the oil and suspended contaminants are removed from the engine.

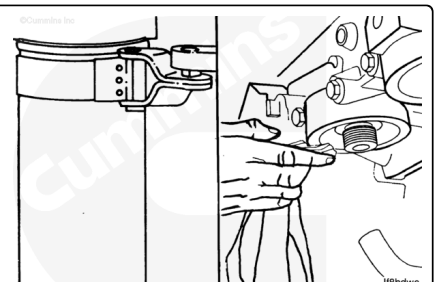
Remove

Clean the area around the lubricating oil filter head and the gasket surface of the filter head.

Use an oil filter wrench, Part Number 3375049, or equivalent, to remove the oil filters.

Be sure the o-ring is removed as it can stick on the filter head.

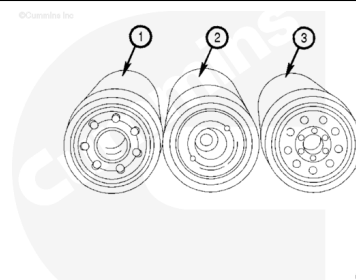
Discard the filters if they are **not** needed for a failure analysis.



Install

⚠ CAUTION ⚠

The bypass and combination filters have the same threads. Verify the correct filter is used for replacement to avoid damage to the engine.



The external appearance of the full-flow (1), the bypass (2), and the combination (3) filters are the same. The accompanying picture identifies the differences among the three filters.

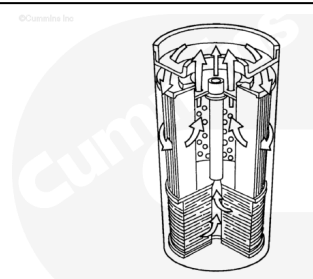
The combination filter contains 1 ½ 12-inch threads. The bypass and combination filters contain 1 ½ 12-inch threads.

A combination oil filter is used on most engines. The upper section of the filter contains the full-flow filter element while the lower section contains the bypass element.

Be sure to use the correct oil filters for the engine.

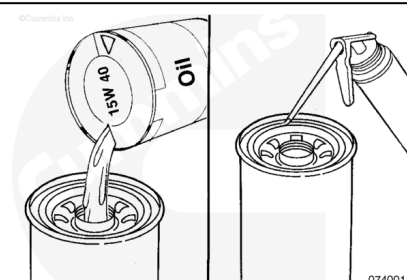
Two combination full-flow/bypass filters are required.

Refer to Section V for oil filter specifications.



⚠ CAUTION ⚠

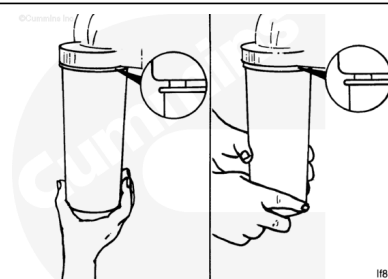
Fill the oil filters with clean lubricating oil. The lack of lubricating during the delay until the filters are pumped full of oil is harmful to the engine.



Apply a light film of lubricating oil to the gasket sealing surface before installing the new filters.

⚠ CAUTION ⚠

Mechanical overtightening can distort the threads or damage the filter element seal.



Install the filter as specified by the filter manufacturer. The tightening instructions are normally printed on the outside of the filter.

Fill

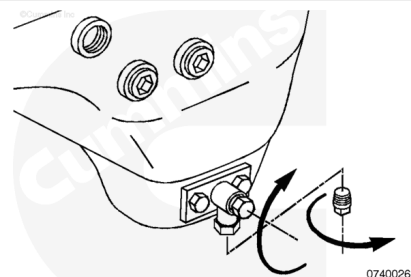
Check and clean the oil drain plug threads and the seal surface.

Close and tighten the oil drain valve.

Torque Value: 68 N•m [50 ft-lb]

Install and tighten the oil drain plug.

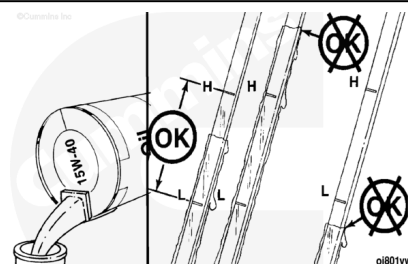
Torque Value: 68 N•m [50 ft-lb]



Use a high-quality 15W-40 multiviscosity oil such as Valvoline® Premium Blue®, or equivalent, in Cummins engines. Choose the correct oil for your operating climate as outlined in Section V of this manual.

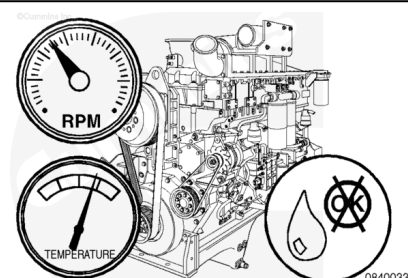


Fill the engine with clean engine oil to the correct level. Total system capacity, including filters, is listed in Section V of this manual.



⚠ CAUTION ⚠

Before starting the engine, complete the steps given in Starting Procedure after Extended Shutdown in Section 1 to be sure the engine receives correct lubrication. Lack of lubrication will damage the engine.



Operate the engine at idle speed to inspect for leaks at the filters and the drain plug.

Turn the engine off.

Wait approximately five minutes to let the oil drain from the upper parts of the engine and check the oil level again.

Add oil as necessary to bring the oil level to the high mark (H) on the dipstick.

